TITLE PAGE

- **Problem Statement ID – SIH26176**
- **Problem Statement Title- ORCA Marine Ecosystem Reasoning with Collaborative Agents**
- **Theme- Disaster Management**
- **PS Category: Software**
- **Team ID- SIH2026-023**
- **Team Name: Team Helios Luna**
- **Team Leader Name : Syed Amaan**
- **Team Leader USN Number: ENG24AM0100**

Sagarvani

Conversational Marine Decision-Intelligence Platform

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Sagarvani is a **conversational marine intelligence platform** that understands a user's question, gathers relevant ocean, weather and geospatial information, and reasons across these sources to produce a **validated recommendation**.

transforms **fragmented marine data** into one coordinated, explainable intelligence layer—**accessible anywhere**, from **smartphones** to **simple helplines** and **low-bandwidth channels.**

The Problem	How ORCA Solves It
Fragmented marine data	Fuses ocean, weather, GIS and risk data through specialized agents.
Data needs manual interpretation	Uses cross-source reasoning to generate actionable recommendations.
Decisions depend on location & context	Combines spatial, weather, ocean and risk factors before recommending.
Limited smartphone/connectivity access	Provides the same intelligence via Web/App, Helpline and Low-Bandwidth.
Conflicting or uncertain results	Validates outputs and triggers re-planning when needed.
Lack of explainability	Provides evidence-backed recommendations with reasoning.

INNOVATION & UNIQUENESS

Multi-Agent Intelligence · Re-planning on Contradictions · Explainable Recommendations · One Intelligence Engine / Three Access Paths

TECHNICAL APPROACH

Methodology and Process of Implementation

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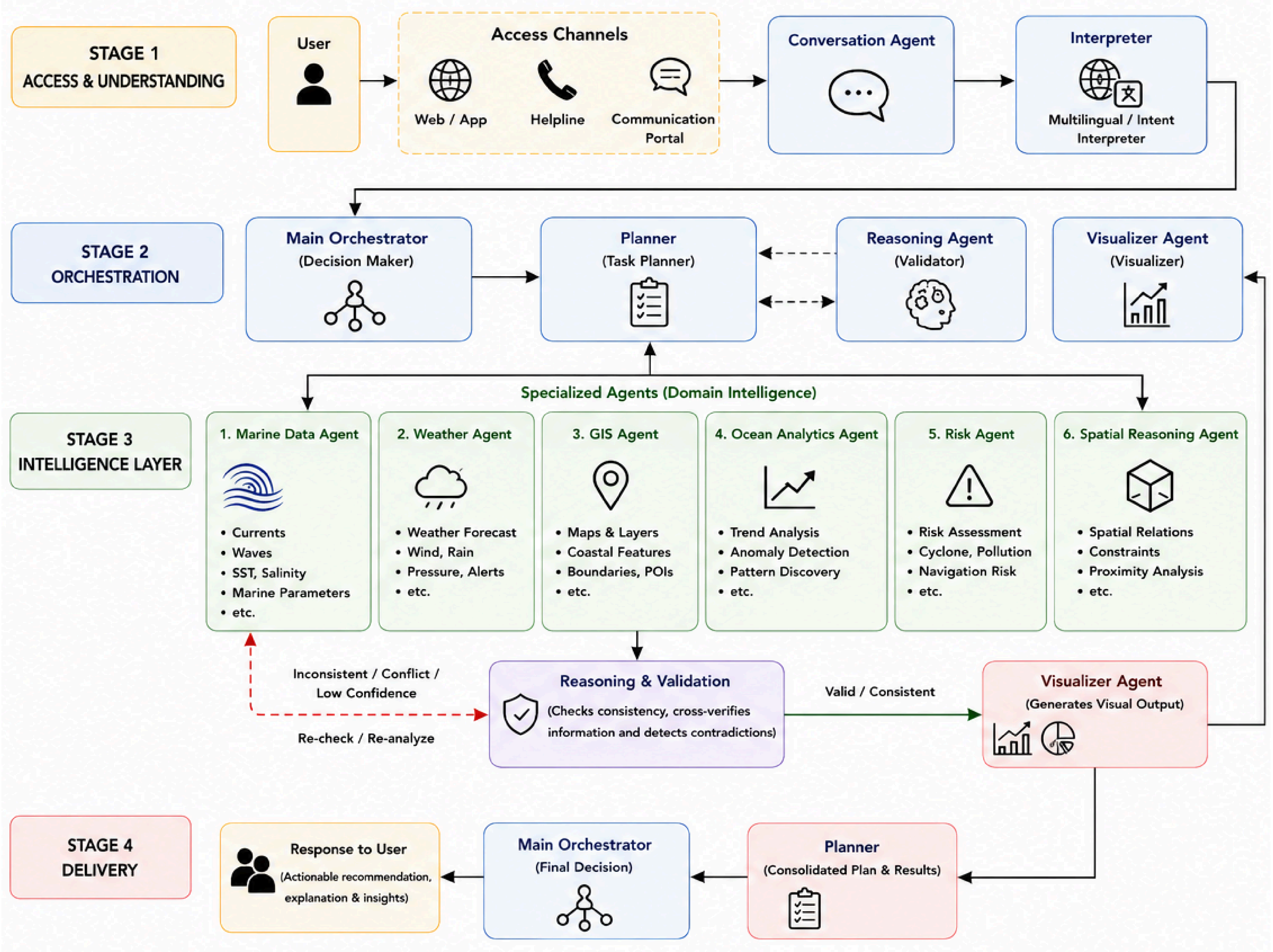
ORCA Process Flow

Understand
↓
Orchestrate
↓
Plan
↓
Retrieve
↓
Analyze
↓
Validate
↓
Visualize
↓
Decide
↓
Deliver



Technologies & Data Sources

- LangChain
- OpenAI
- Python
- PostgreSQL
- GeoPandas
- Leaflet
- Copernicus
- INCOIS
- meteoblue
- Streamlit
- twilio
- Docker
- Redis
- FastAPI
- bhashini (AI for Bharat)
- sarvam
- BHUNIDHI (Geospatial Data Repository)
- MOSDAC (Digitalising the Ocean)
- VCSS (Vessel Communication Support System)

FEASIBILITY AND VIABILITY

1. FEASIBILITY

Can We Build It?

- Existing Data – Marine, weather & GIS datasets are available through established platforms/APIs.
- Mature Technology – LLMs, GIS and multi-agent frameworks are ready for implementation.
- Modular Architecture – Start with core agents and limited datasets, then expand.

2. CHALLENGES

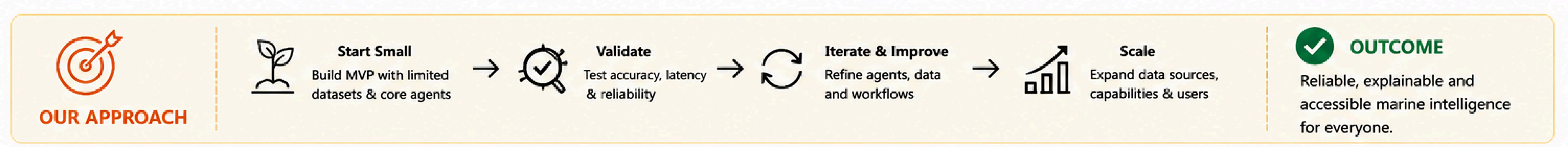
What Can Go Wrong?

- Data Heterogeneity – Different formats, quality and update cycles.
- API Dependency – Rate limits, downtime and source changes.
- AI Reliability – Risk of incorrect or hallucinate recommendations.
- Connectivity – Poor network availability in coastal/remote areas.

3. MITIGATION READY

How We Handle It

- Data Validation → Check quality, completeness and consistency.
- Fallback Sources → Switch to alternate APIs/data sources when unavailable.
- Parallel Execution → Run independent agent tasks concurrently.
- Validation + Re-checking → Detect conflicts before final output.



IMPACT AND BENEFITS

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IMPACTS ON TARGET AUDIENCE

1. FISHERMEN

Safer & simpler decisions

- Marine & weather information
- Local-language access
- Helpline / keypad-phone support
- Safer fishing & trip planning

2. RESEARCHERS

Faster marine analysis

- Integrated datasets
- Advanced visualization
- Faster data exploration
- Scenario-based analysis

3. MARITIME OPERATORS

Smarter operational planning

- Weather & ocean intelligence
- Route / operational planning
- Risk assessment
- Context-aware recommendations

BENEFITS OF THE SOLUTION

1. SOCIAL

Improved Marine Safety & Accessibility

- Wider access to marine intelligence
- Support for non-smartphone users
- Safer coastal operations

2.ECONOMICAL

Better Operational Decisions

- Faster evidence-backed decisions
- Improved fishing & maritime operations
- Better utilization of marine resources

3.ENVIRONMENTAL

Stronger Coastal Resilience

- Better risk awareness
- Improved disaster preparedness
- Support for sustainable marine activity

RESEARCH AND REFERENCES

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Sources	Data Utilized
INCOIS	PFZ Advisories; Sea Surface Temperature (SST); Chlorophyll; Wave Height; Wave Period; Ocean Currents; Wind; Sea State Forecasts; Tsunami & High-Wave Warnings
IMD	Fishermen Warnings; Cyclone Information; Wind & Rainfall Forecasts; Coastal Weather Bulletins; Sea Area Bulletins; Port Warnings
ISRO Bhuvan / NRSC	Satellite Imagery; PFZ Maps; Land & Coastal Layers; Elevation; Hydrology; Disaster & Geospatial Data
Bhoonidhi – ISRO/NRSC	Earth Observation Satellite Data; Remote-Sensing Imagery; Satellite-Derived Geospatial Products
MOSDAC – ISRO/SAC	Meteorological & Oceanographic Satellite Data; SST; Chlorophyll; Ocean Currents; Wind; Cyclone & Weather Information
BHASHINI + Sarvam AI	Speech-to-Text; Text-to-Speech; Multilingual Translation; Transliteration; Indian-Language Voice & Text Processing